



Introduction to Computer Networks

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About the Course

- It covers the fundamentals of how devices communicate over a network, including hardware, software, and protocols.
- It explores various network topologies, their design, and how to optimize network performance and reliability.
- Students learn about services and protocols of each layer of the standard TCP/IP protocol suite



Prerequisites

- Fundamentals of computer science
- Data structure and algorithms
- Discrete mathematics (Logic, Set theory, Graph theory)
- Operating Systems



Books

- Text

- Behrouz A. Forouzan (2022), Data communication and Networking with TCPIP Protocol Suite, 6th Edition, McGraw-Hill, India.

- Reference

- A. S. Tanenbaum David J. Wetherall, and Nick Feamster (2022), Computer Networks, 6th edition, Pearson Education/ PHI, New Delhi, India.
- J. F. Kurose, and K. W. Ross (2017), Computer Networking: A top-down approach, 6th edition, Pearson Education, India.



Evaluation Criteria

- Mid Sem – 30 Marks
- End Sem – 50 Marks
- Teacher Assessment – 20 Marks
 - Classroom Assignment
 - Technical Report / Survey on Challenges in various domain compute networks
 - Attendance



Data Communication

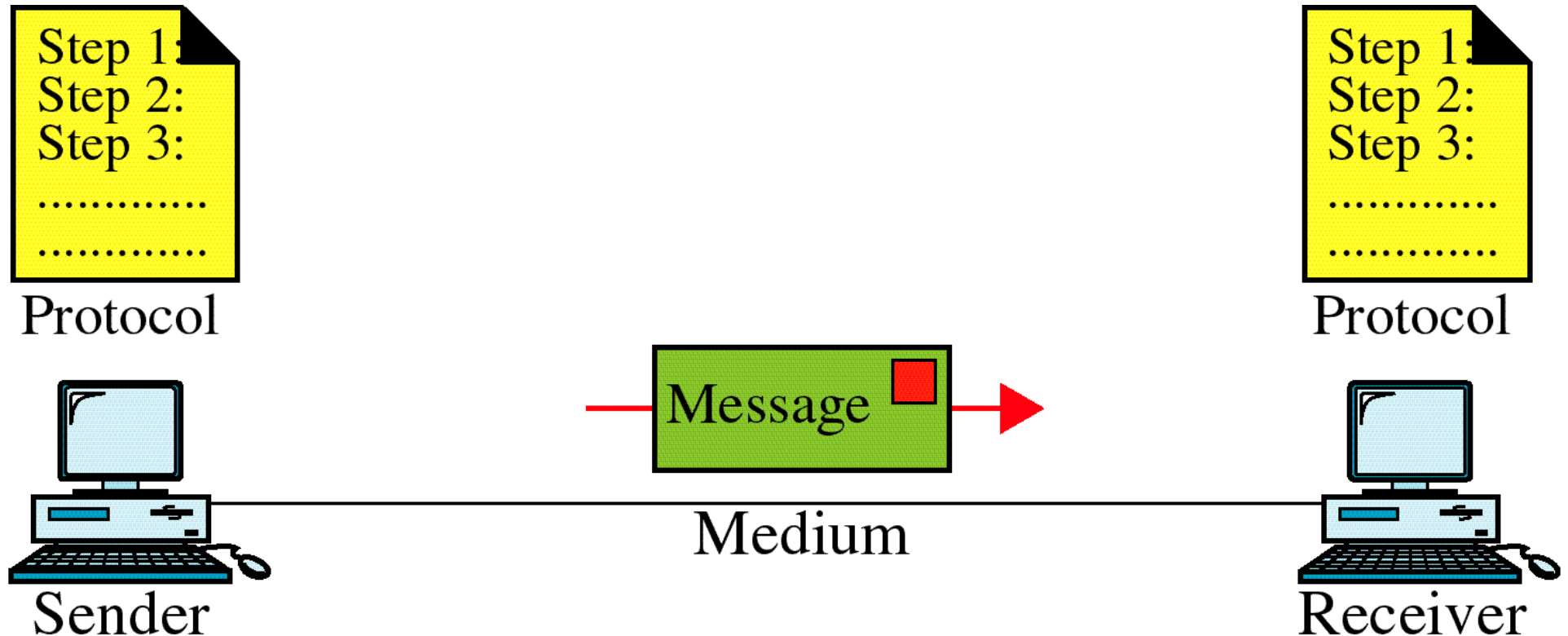
- It is the **exchange** of data between two **devices** via a transmission **medium**
- For data communications to occur, the communicating devices must be **part** of a **communication system**
- Communication system is a combination of **hardware** and **software**.



Effectiveness of Data Communication

- **Delivery**
 - The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
- **Accuracy**
 - The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
- **Timeliness**
 - The system must deliver data in a timely manner.
 - Data delivered late are useless
 - Video and audio must be delivered as they are produced, in the same order without significant delay(real-time transmission)
- **Jitter**
 - The variation in the packet arrival time
 - It is the uneven delay in the delivery of audio or video packets

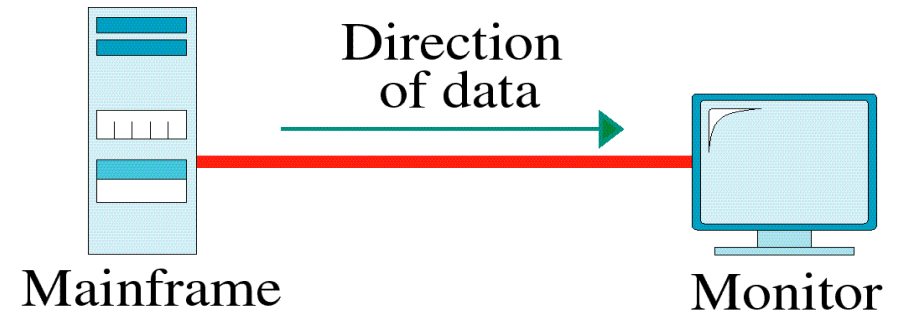
Components of Data Communications



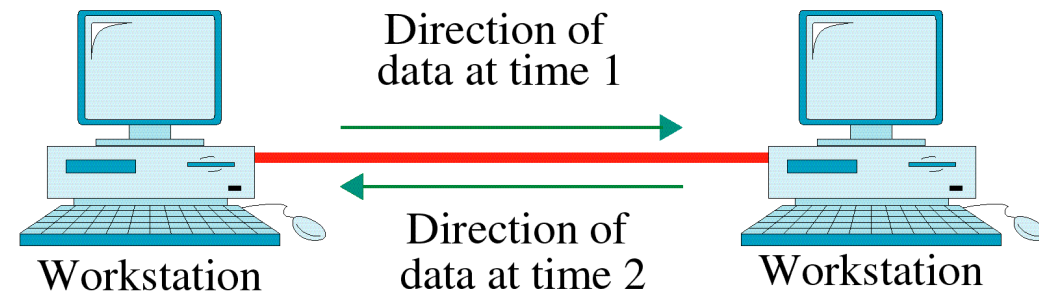


Data Transmission Modes/Data Flow

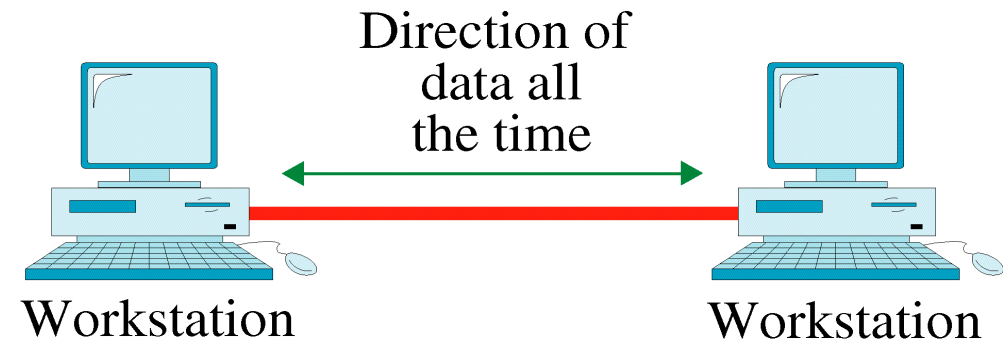
Simplex



Half Duplex



Full Duplex





Network Criteria

- Performance
 - **Transit time** : the amount of time required for a message to travel from one device to another
 - **Response time** : the elapsed time between an inquiry and a response
 - **Performance** typically depends on : **the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software.**
 - Performance is often **evaluated** by : **throughput** and **delay**
 - Throughput is the number of packets transmitted per second



Network Criteria(2)

- Reliability
 - The frequency of failure
 - The time it takes a link to recover from a failure
 - The network's robustness in a catastrophe.
- Security
 - Protecting data from unauthorized access
 - Protecting data from damage
 - Development, and implementing policies and procedures for recovery from breaches and data losses



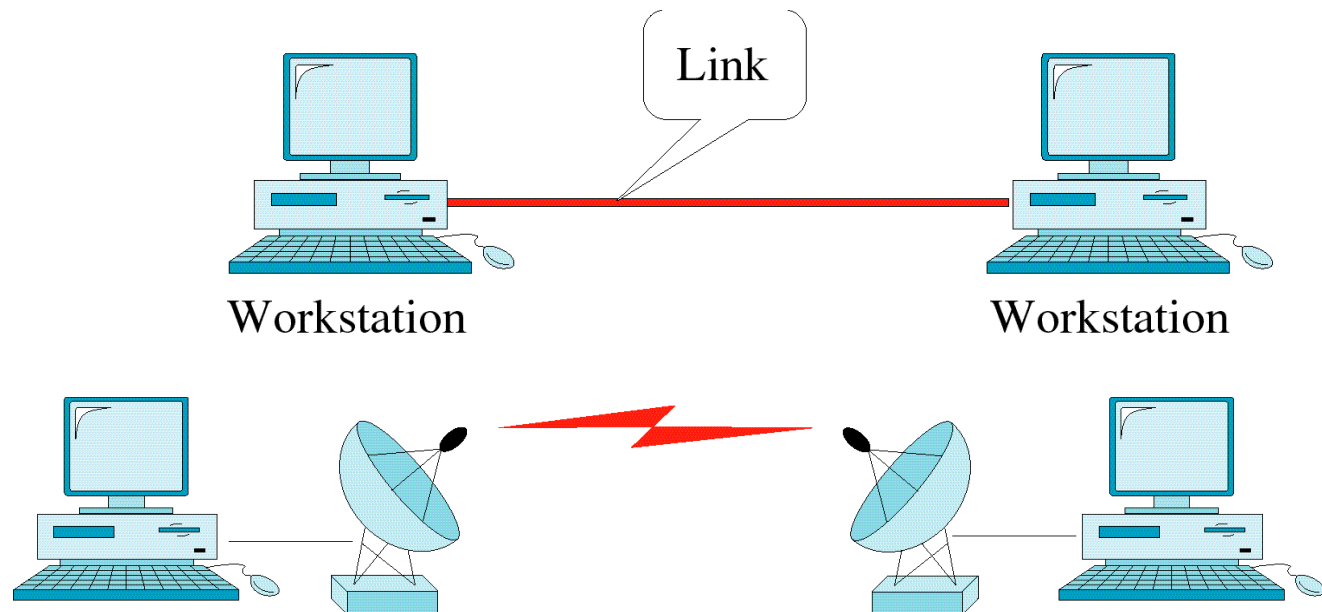
Type of Connection

- Two or more devices are connected through **links**. (Denoted as a line)
- For communication two devices must be connected in some way to the same link at the same time.
- Two possible types
 - Point-to-point
 - Multipoint

Type of Connection(2)

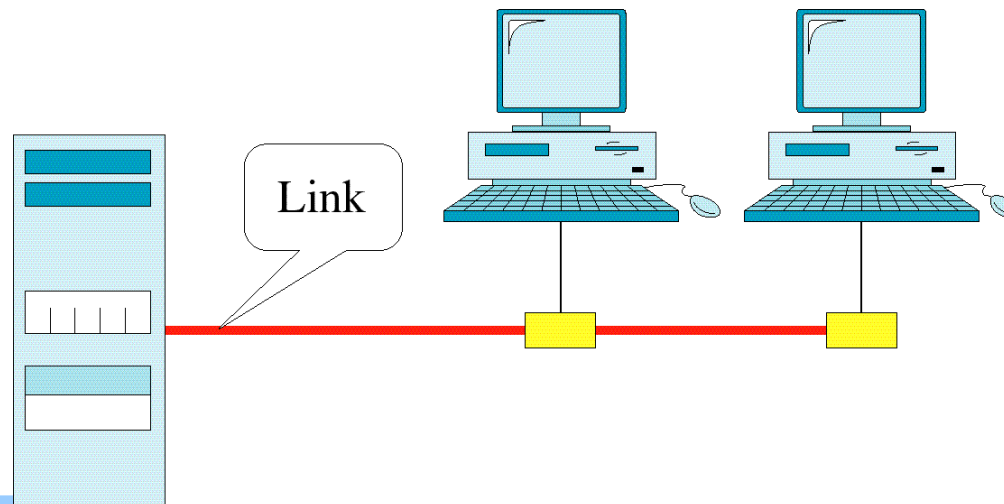
- **Point-to-Point**

- Provides a dedicated link between two devices
- The entire capacity of the link is reserved for transmission between those two devices
- Mostly use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible
- Example : Infrared communication using TV remote is point-to-point



Type of Connection(2)

- Multipoint
 - More than two specific devices share a single link
 - The capacity of the channel is shared, either spatially or temporally
 - If several devices can use the link simultaneously, it is a spatially shared connection
 - If users must take turns, it is a timeshared connection



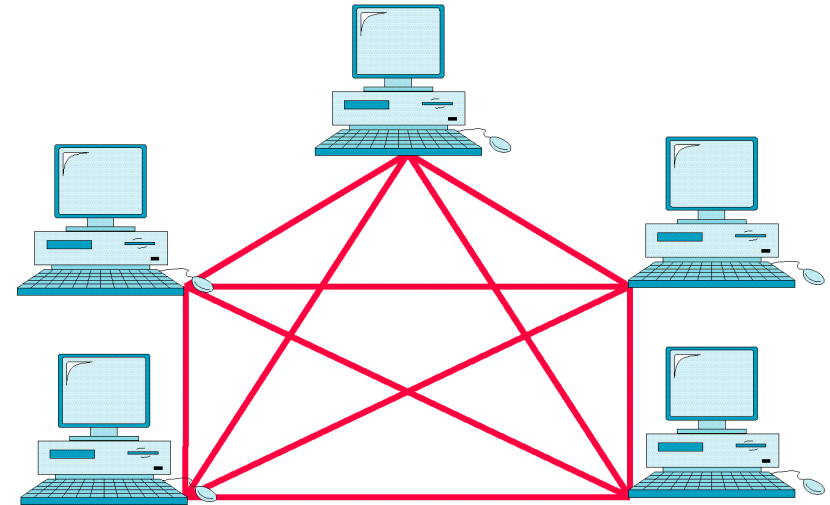


Network Topology

- The geometric representation of the relationship of all the links and linking devices (a.k.a nodes) to one another
- Physical topology : How the link cables run
- Logical topology : How a message travels in the network
- 4 basic types : Mesh, Star, Bus, and Ring
- Other forms : Tree and Hybrid

Mesh Topology

- A **dedicated point-to-point** link to **every** device
- For 'n' number of nodes, number of full duplex links required is $n(n-1)/2$
- Every device on the network must have **n - 1 input/output ports**
- **Advantages**
 - Dedicated links eliminates the traffic problems
 - A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system
 - Privacy or security
 - Fault identification and fault isolation is easy
- Example : connection of telephone regional offices



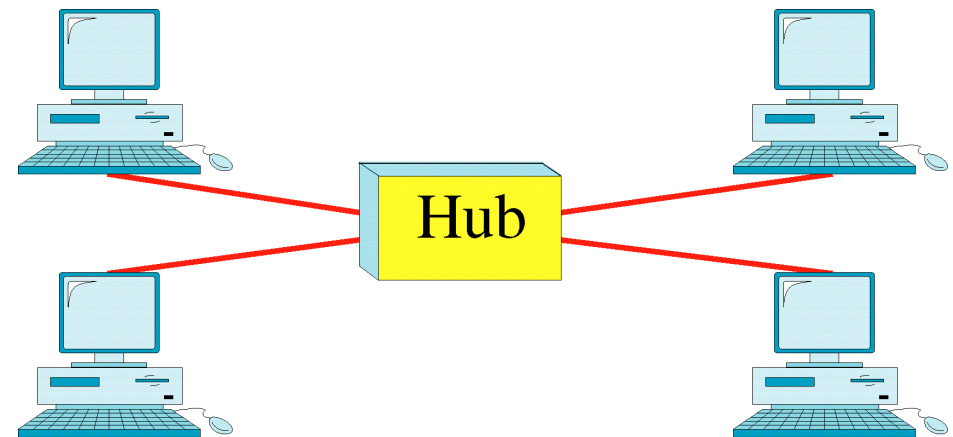
- **Disadvantages**
 - Amount of cabling and the number of I/O ports required
 - Installation and reconnection are difficult
 - The sheer bulk of the wiring can be greater than the available space
 - Hardware cost may be expensive
- Usually implemented in a **backbone** of the network

Star Topology

- Each device has a dedicated P2P link only to a central controller, usually called a hub
- The devices are not directly linked to one another
- The controller acts as an exchange

- **Advantages**

- A star topology is less expensive than a mesh topology
- Each device needs only one link and one I/O port to connect
- Robustness - If one link fails, only that link is affected
- Easy fault identification and fault isolation

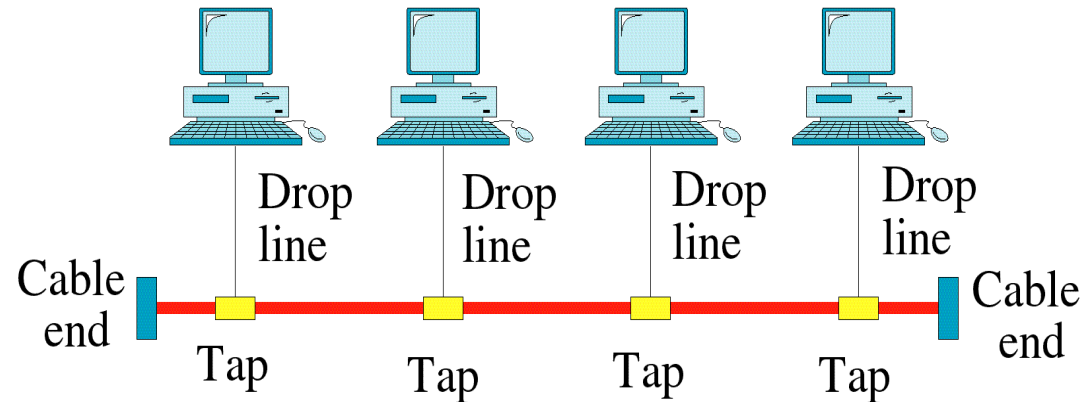


- **Disadvantages**

- The dependency of the whole topology on one single point, the hub
- Prone to **single point of failure**
- More cabling is required in a star than in ring or bus
- Commonly used in local-area networks (LANs)
- High-speed LANs often use a star topology with a central hub

Bus Topology

- Oldest topology
- Uses multipoint connection
- One long cable acts as a backbone
- Devices are connected using drop lines and taps
- **Advantages**
 - Easy installation
 - Less cable required compared to mesh and star

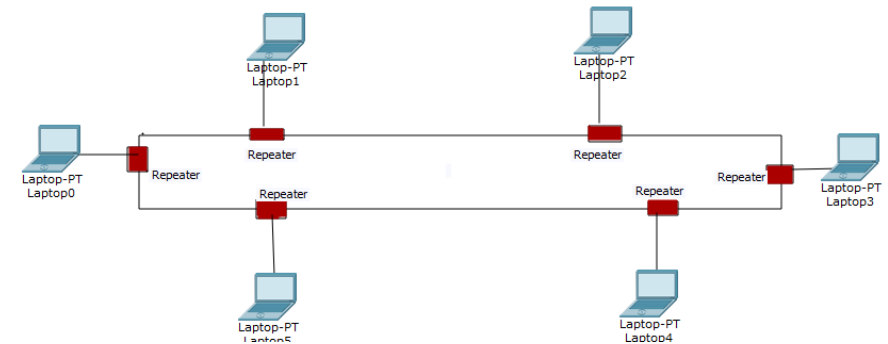


- **Disadvantages**

- Difficult in fault detection and isolation
- Reconfiguration is not possible
- Adding new device requires a new main bus
- Bus act as single point of failure

Ring Topology

- Developed by IBM
- Each device has a dedicated point-to-point connection with only the two devices on either side of it
- A signal is passed along the ring in **one direction**, until it reaches its destination. Call **token passing** mechanism
- Each device incorporates a **repeater** to regenerate the bits and passes them along



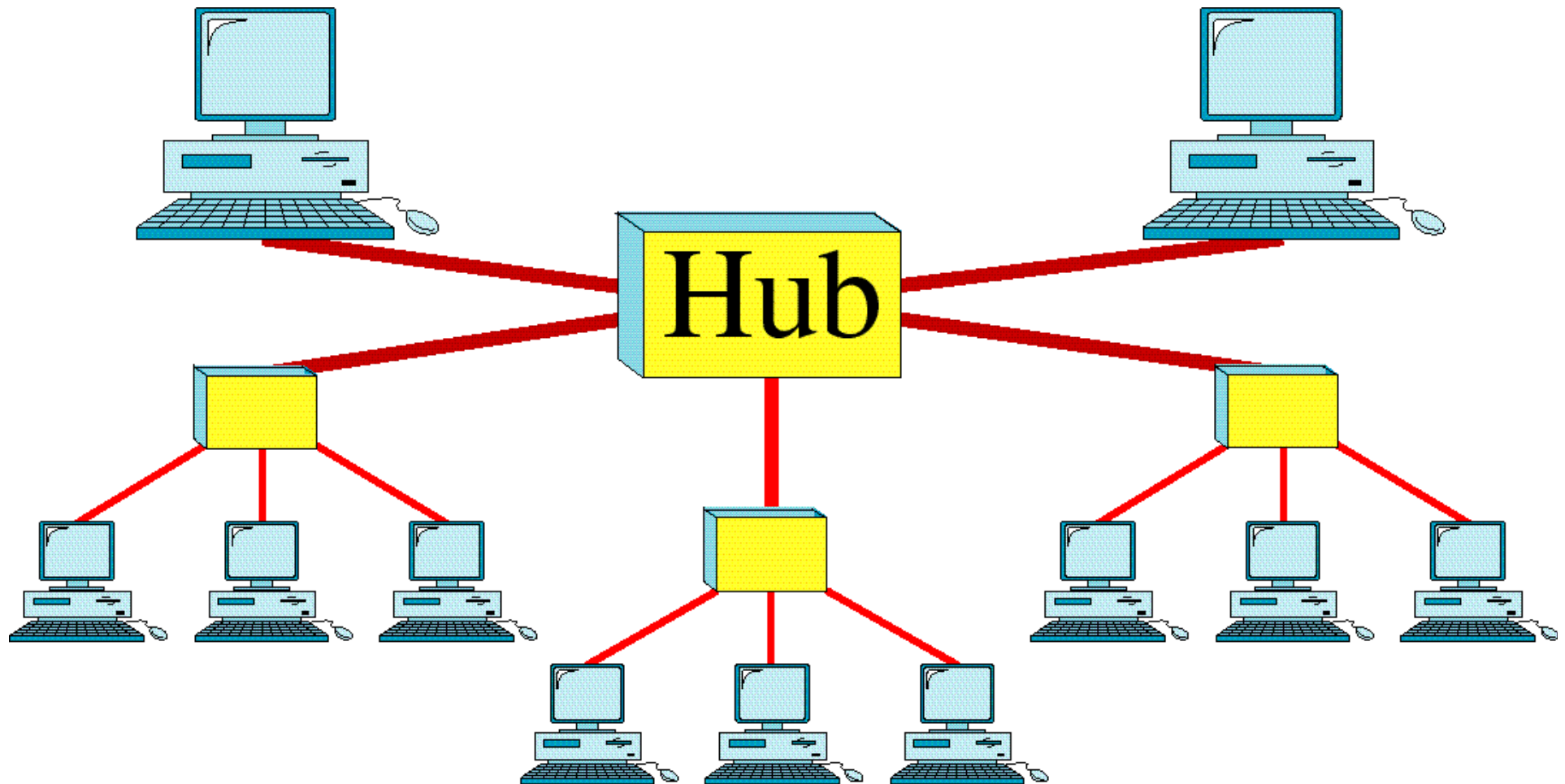
- **Advantages**

- Relatively easy to install and reconfigure
- Fault identification is easy

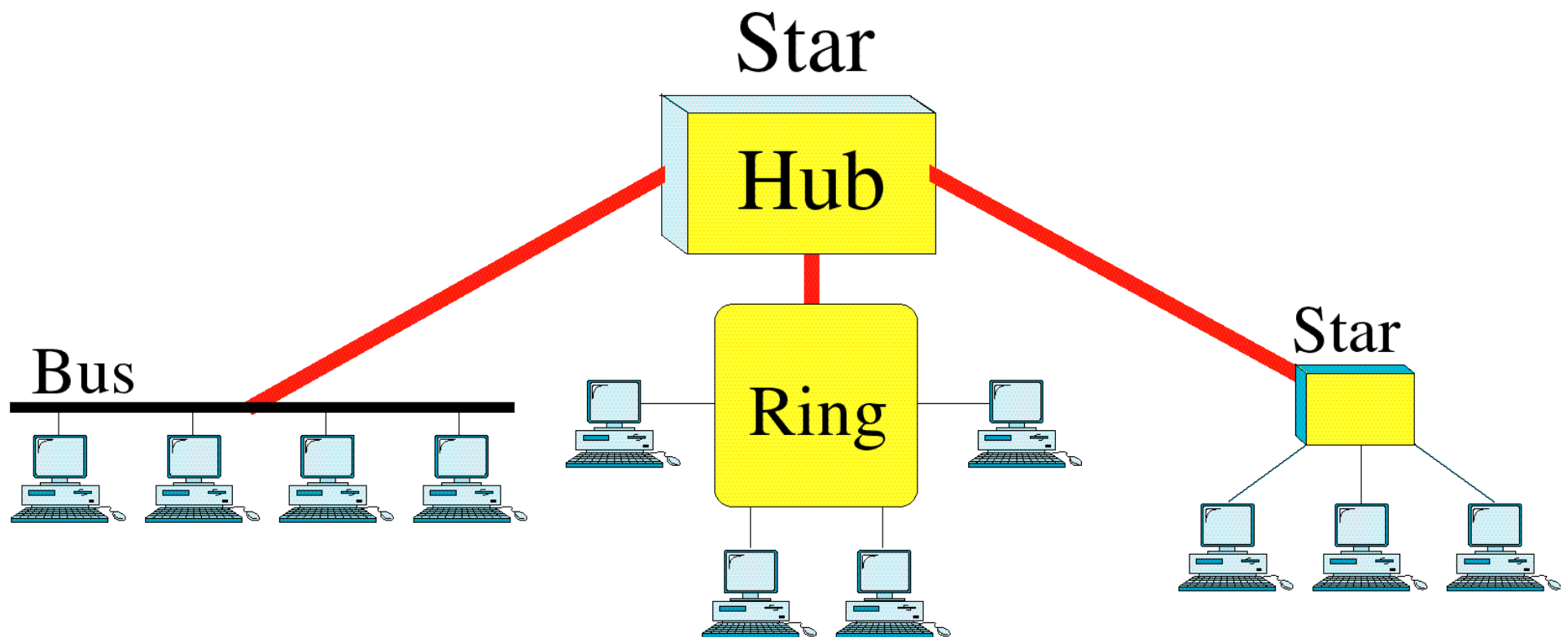
- **Disadvantages**

- Unidirectional traffic
- A fault in one link may disable the entire network

Tree Topology



Hybrid Topology



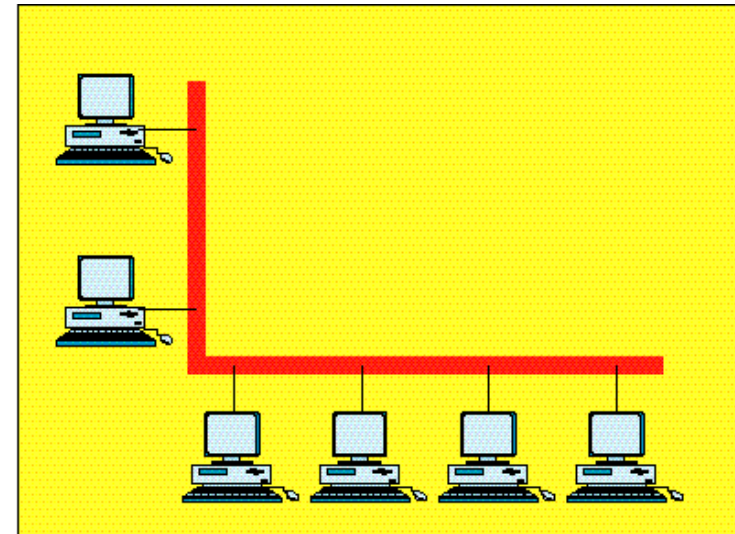


Categories of Networks

- Criteria to distinguishing one type of network from another are
 - Size
 - Geographical coverage
 - Ownership
- Local Area Networks (LANs)
 - Meant for Short distances
 - Designed to provide local interconnectivity
- Wide Area Networks (WANs)
 - Long distances
 - Provide connectivity over large areas
- Metropolitan Area Networks (MANs)
 - Provide connectivity over areas such as a city, a campus

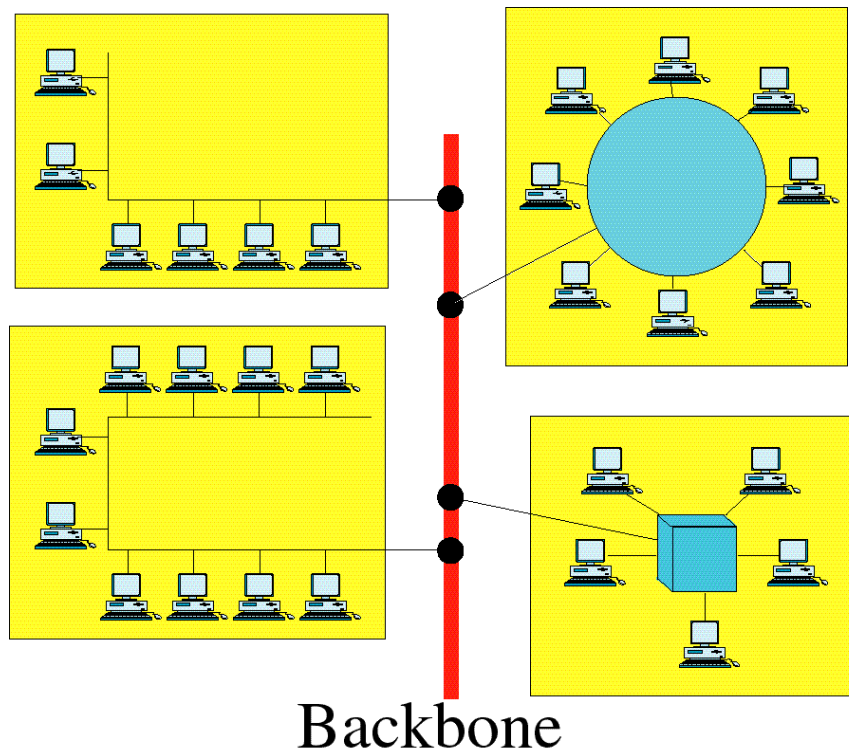
Local Area Networks (LANs)

- Privately owned
- Links devices in the same office, building, or campus.
- Each host has an identifier, an address, that uniquely defines the host.
- Size is limited to a few kilometers.
- Allow resources to be shared (hardware, software, or data).
- It does not define the minimum or maximum number of hosts.
- It use either a common cable or a switch
- One of the computers may be given a large capacity disk drive and may become a server to clients



Single building LAN

Local Area Networks (LANs)

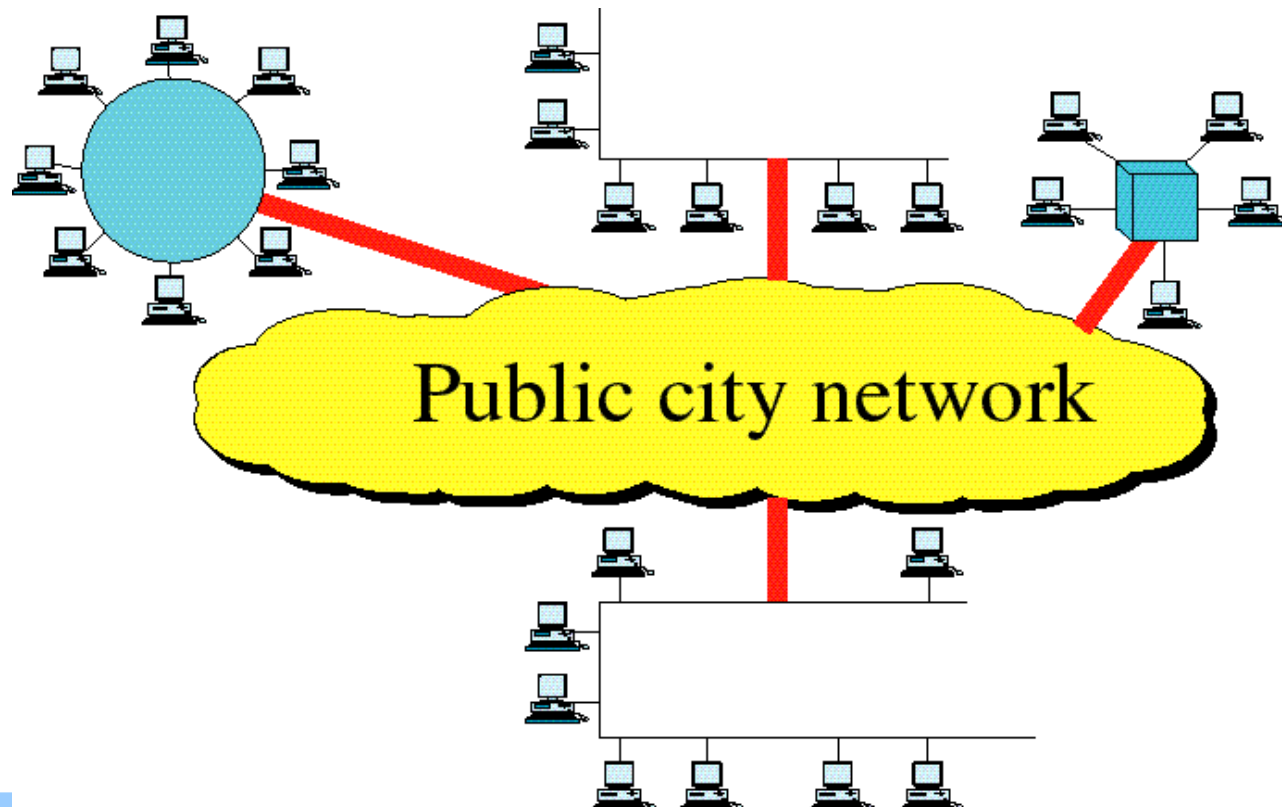


Multiple building LAN

- Software can be stored on this central server and used as needed by the whole group
- LAN may be determined by licensing restrictions on the number of users per copy of software, or by restrictions on the number of users licensed to access the operating system.
- LANs are distinguished from other types of networks by their transmission media and topology
- A given LAN will use only one type of transmission medium
- The most common LAN topologies are bus, ring, and star.
- Today, however, speeds are normally 100 or 1000 Mbps.

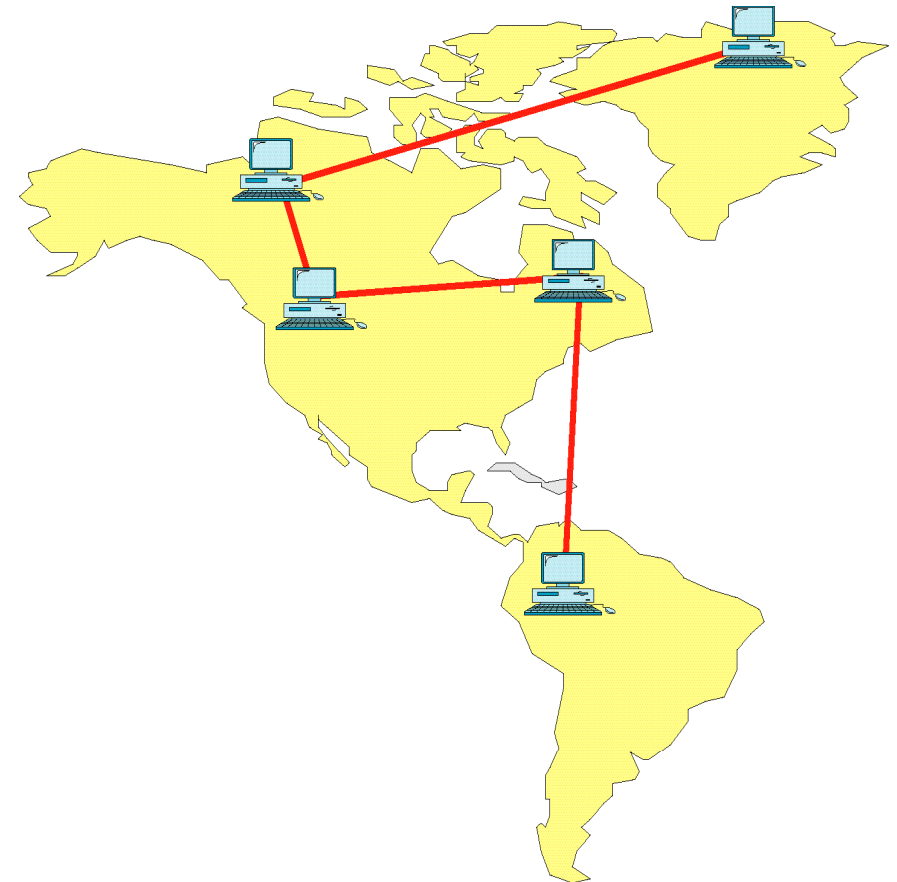
Metropolitan Area Networks (MANs)

- Collection of LANs with the same geographical area, for instance a city.
- Often acts as a high speed network to allow sharing of regional resources.
- Companies that have several branches within a city such as banks, might find a MAN useful to them.
- Example : Telephone company network, cable TV network



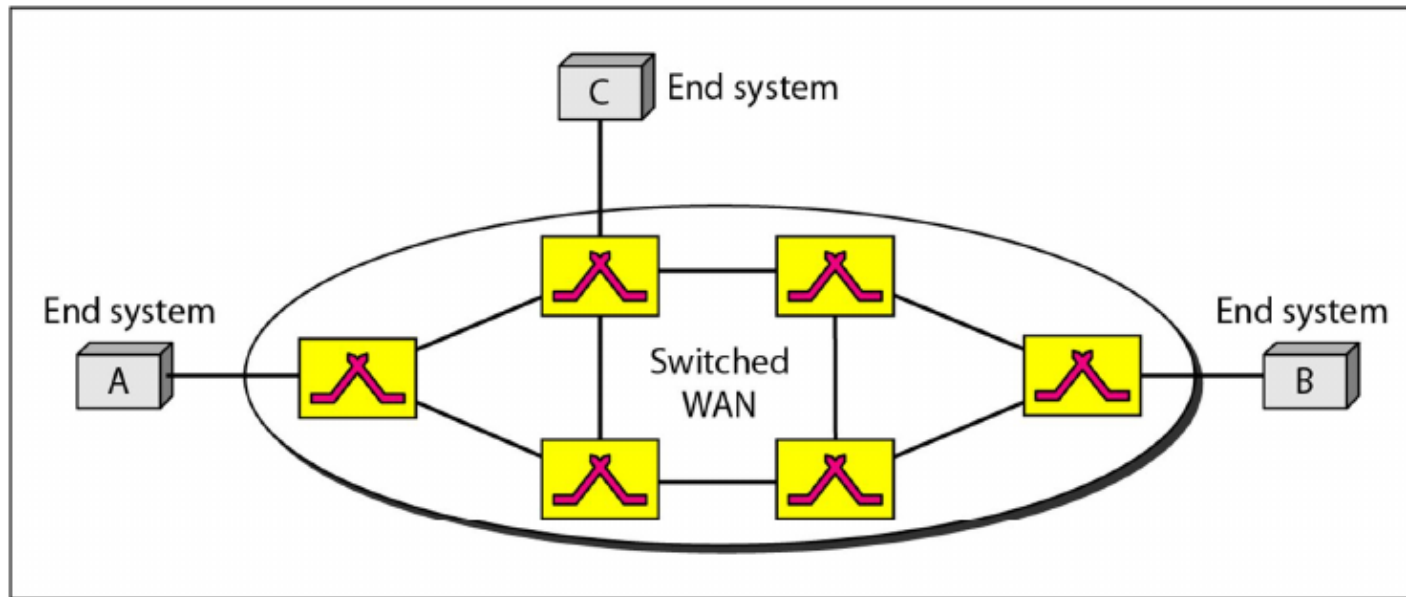
Wide Area Networks (WANs)

- Provides **long-distance transmission** of data, image, audio, and video information over large geographic areas that may comprise a **country**, a **continent**, or even the whole world
- Can be
 - The backbones that connect the Internet (a switched WAN)
 - As simple as a dial-up line that connects a home computer to the Internet (point-to-point WAN)



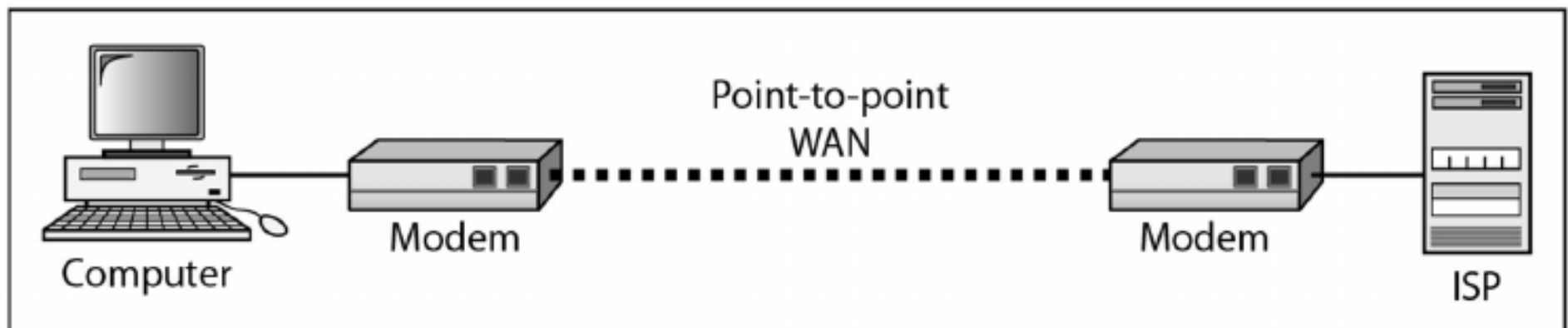
Switched WAN

- Connects the end systems, which usually comprise a router that connects to another LAN or WAN
- Example: X.25, asynchronous transfer mode (ATM) network, Wireless LAN

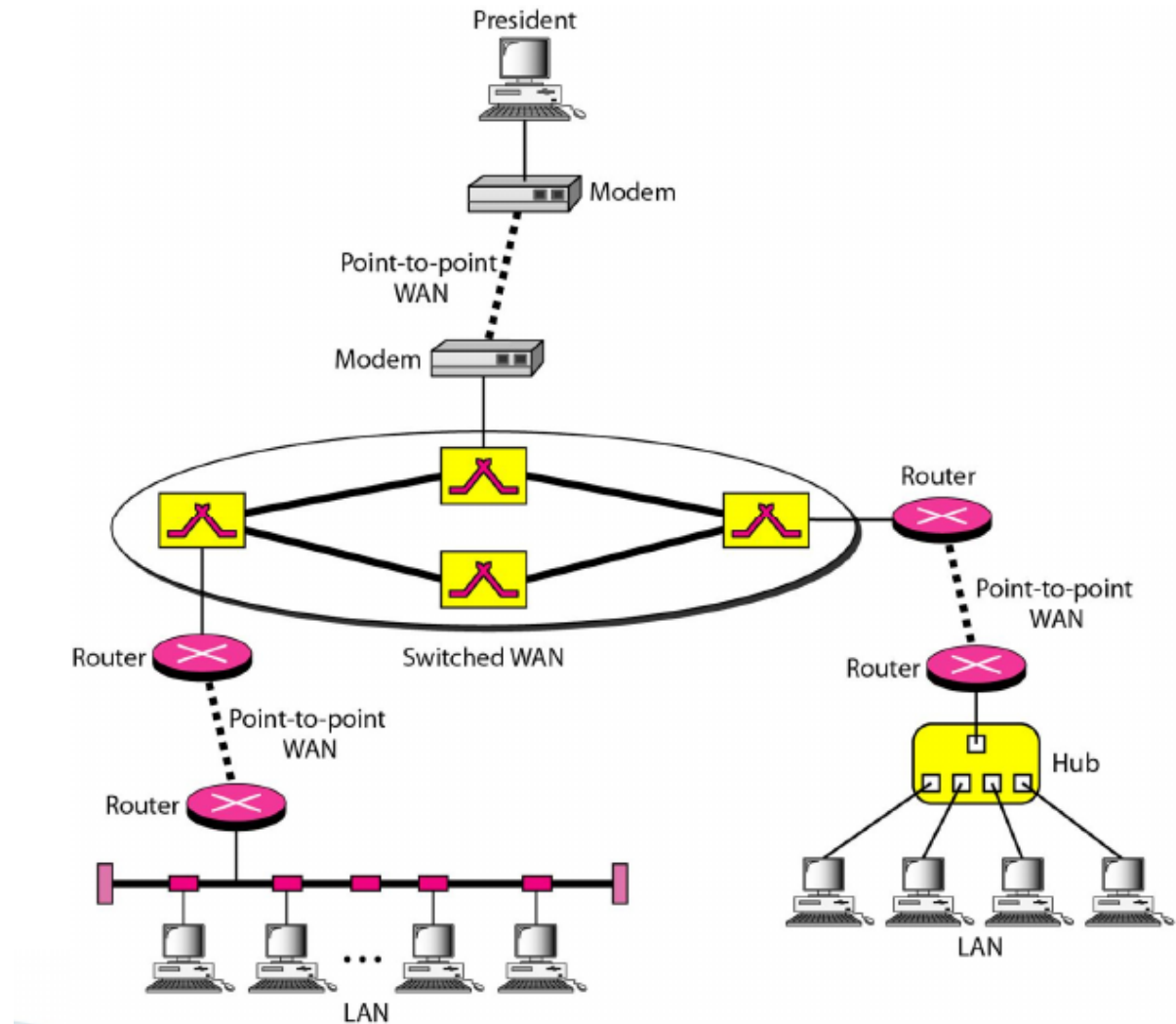


Point-to-point WAN

- A line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP)
- Often used to provide Internet access

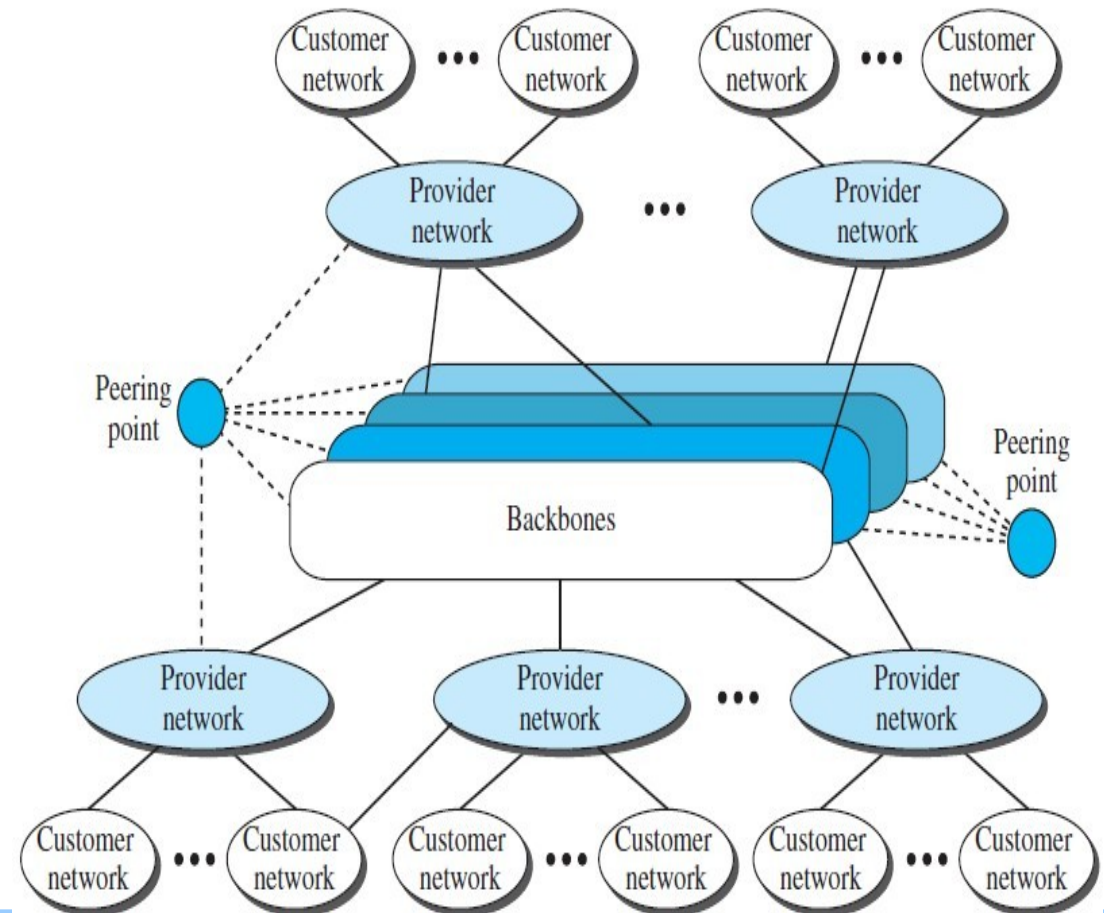


An Internetwork (internet)



The Internet

- It is **composed of thousands of interconnected** networks.
- The Internet is a structured, organized system





Protocols

- A protocol is a set of rules that govern data communications
- A protocol defines what is communicated, how it is communicated, and when it is communicated
- Key elements
 - **Syntax** - refers to the structure or format of the data, meaning the order in which they are presented
 - **Semantics** - refers to the meaning of each section of bits and what action is to be taken based on that interpretation
 - **Timing** - when data should be sent and how fast they can be sent



Standards

- Creating and maintaining an open and competitive market for equipment manufacturers
- National and international interoperability of data and telecommunications technology
- Standards provide **guidelines** to manufacturers, vendors, government agencies, and other service providers
- Two categories
- **De facto** (meaning "by fact" or "by convention")
 - Not been approved by an organized body but have been adopted as standards through widespread use are de facto standards
- **De jure** (meaning "by law" or "by regulation").
 - Legislated by an officially recognized body are de jure standards



Standards Creation Committees

- International Organization for Standardization (ISO)
- International Telecommunication Union-Telecommunication Standards Sector (ITU-T)
- American National Standards Institute (ANSI)
- Institute of Electrical and Electronics Engineers (IEEE)
- Electronic Industries Association (EIA)
- Internet Standards -
 - Internet draft
 - Request for Comment (RFC)



Next : Network Models